

PEER REPAIR AND FORECAST FUSION FOR MISSING SENSOR HISTORIES

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ABSTRACT

Forecasting after a sensor outage requires deciding how missing recent values should interact with the longer observed history. We present a fixed procedure that repairs recent target gaps using regressions fitted on observed local and related-series prefixes, retains the available long context of a frozen Chronos-2 forecaster, and equally combines its point prediction with a prefix-fitted vector autoregression. The procedure preserves observed and auxiliary entries and requires one foundation-model forecast per origin. On 45 natural-outage windows from 12 Beijing air-quality stations, it attains standardized MAE/MSE of 0.388053/0.244688. Relative to native prediction using the same long history and VAR fusion, the reductions are 5.98%/7.26%; relative to long native prediction alone, they are 19.96%/31.90%. Gaussian repair with the same fusion performs nearly identically, while other controls are stronger under artificial outages and at horizon 96. Available Gaussian and fusion components also fail to improve natural-gap forecasting on an HDB parking cohort. These developmental results identify an effective fixed procedure for the studied natural sensor outages and delimit its current evidence across missingness conditions and applications.

1 INTRODUCTION

Air-quality monitoring networks must forecast pollutant concentrations when a target sensor stops reporting. Consider a station whose PM2.5 and PM10 measurements have been absent for the last day. Its earlier history, local meteorological measurements and contemporaneous readings from other stations may remain available. A pretrained forecaster can consume this incomplete history directly, while a repair procedure can infer the missing target values before prediction. The useful intervention is the one that improves the future forecast under the information available at the forecast origin.

Existing methods provide several relevant capabilities. Neural imputers such as SAITS and TimeMixer++ learn temporal and multivariate representations, and MoTM transfers pretrained imputation representations (Du et al., 2023; Wang et al., 2025; Le Naour et al., 2025). Imputation quality and downstream prediction quality can differ (Le Morvan et al., 2021), and downstream evaluation is already included in time-series imputation benchmarks (Du et al., 2024). Chronos-2 can share information across related series and accept multivariate histories (Ansari et al., 2025). These capabilities leave a concrete deployment choice: how much of an incomplete history to repair, which observations to preserve, and how to use a statistical forecast alongside the frozen model.

We study a fixed procedure for forecasting after target-sensor outages. Its input includes an observed historical prefix for fitting simple statistical models and a longer, possibly incomplete history ending at the forecast origin. The procedure repairs only recent missing target entries, retains the available long history, and combines the frozen forecaster’s point prediction with a statistical forecast. The target is downstream MAE, with MSE reported throughout. The forecasting backbone is unchanged, and current hidden measurements and future outcomes are excluded from the procedure.

Two difficulties motivate this construction. First, unavailable target values must be inferred from a changing set of observed variables. Replacing all variables can also alter useful auxiliary evidence, while restricting the forecaster to a short repaired window discards older observations. Second, a plausible repaired trajectory can still induce an inaccurate forecast. A separate statistical prediction describes another use of the same pre-origin evidence, but combining predictions helps only under suitable component errors. Both effects require matched long-history and statistical-fusion controls.

We define Peer Repair and Fusion, which uses prefix-fitted ridge regression on available local and peer observations, inserts the estimated target values into a protected long history, and equally combines the resulting Chronos-2 forecast with a prefix-fitted vector autoregression (VAR). In the motivating PM example, reporting peers estimate the unavailable recent PM values; older target measurements remain in the model context. The VAR branch filters the observed recent history and extrapolates its final state. Both branches end at the same forecast origin, and the fusion weight is fixed across evaluation cases.

The paper makes three concrete contributions. Section 4 specifies an executable repair-and-fusion procedure with explicit fitting, observation-preservation and information boundaries. Section 4.4 characterizes its preserved inputs and the standard squared-error condition governing fusion. Section 5 evaluates one fixed method across natural outages, artificial target outages and a longer forecasting horizon, with additional component-transfer results on parking availability. On 45 natural-outage windows from 12 Beijing stations, the method obtains MAE/MSE of 0.388053/0.244688, improving on the matched long-native-plus-VAR control by 5.98%/7.26%. The other panels delimit this result: different fixed controls are stronger on artificial outages and at horizon 96. The procedure and application focus are selected from development evidence, whose interpretation is made explicit in the experimental design.

2 RELATED WORK

Imputation and downstream prediction. Le Morvan et al. (2021) analyze the relationship between imputation and prediction with missing covariates. SAITS uses self-attention for imputation (Du et al., 2023); TimeMixer++ provides representations for several time-series tasks, including imputation (Wang et al., 2025); and MoTM uses continuous modeling for pretrained time-series imputation (Le Naour et al., 2025). TSI-Bench evaluates imputation methods under multiple missingness conditions and includes downstream tasks (Du et al., 2024). Our evaluation fixes the forecasting backbone and separately controls the repair region, the forecaster’s history and the auxiliary variables it receives.

Missingness-aware and pretrained forecasting. Chronos-2 uses group attention to exchange information among related series, variables and covariates (Ansari et al., 2025). S4M incorporates missingness representations into a trained forecasting architecture (Peng et al., 2025). Our procedure uses an existing Chronos-2 checkpoint and fits only external statistical models. A low-rank adaptation control (Hu et al., 2021) tests whether updating a small set of additional parameters provides a stronger comparison. Related-series conditioning and adaptation are existing capabilities; neither is treated as an original primitive here.

Statistical repair and forecast combination. Ridge regression stabilizes linear estimation with correlated predictors (Hoerl & Kennard, 1970). Linear forecast combination has a long history (Bates & Granger, 1969), and the squared-error decomposition for averaged predictions is established (Krogh & Vedelsby, 1994). We use these components to define a specific target-repair procedure for a frozen multivariate forecaster. The empirical question is whether this fixed combination adds value beyond the same long history, the same VAR branch, and strong alternative repairs. We retain those comparisons when their performance is similar to or better than the proposed procedure.

3 PROBLEM SETTING

Let t be the forecast origin, D the number of available local and peer variables, and $\mathcal{T}$ the set of K target variables. The history $O_{<t}$ contains observed values and missing entries, with mask $M_{uj} = 1$ exactly when variable j was observed at time $u < t$. An earlier fitting prefix ends at $t_0 < t$. For each variable, its observed-prefix mean μ_j and population standard deviation s_j define $z_{uj} = (O_{uj} - \mu_j)/s_j$. A scale at most 10^{-12} is replaced by one. Missing entries remain missing after this transformation.

The frozen point-forecasting interface F_H returns the 0.5-quantile prediction for the next H steps. The procedure may fit statistical parameters on the prefix and read observations before t . It cannot use current artificially hidden values, measured future peer values or the future targets $Y \in \mathbb{R}^{H \times K}$.

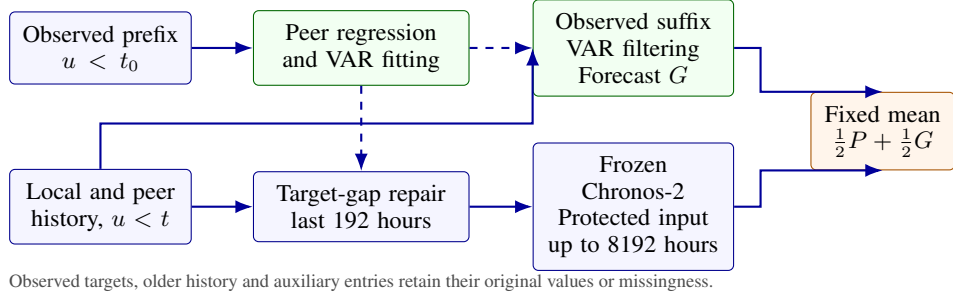


Figure 1: Peer Repair and Fusion. Prefix observations fit the peer regressions and VAR. At prediction time, regressions fill only missing target cells in the most recent 192 hours. Earlier history and auxiliary entries retain their original values or missingness. The repaired long history supplies one frozen Chronos-2 forecast. A second branch filters the unrepaired recent observations with the VAR and forecasts its final state. The two point predictions receive fixed equal weights. All input observations precede the forecast origin.

A future observation mask V is available only for scoring. This setting permits target-prefix statistical fitting and therefore does not describe a completely training-free forecasting pipeline.

4 PEER REPAIR AND FORECAST FUSION

Figure 1 shows the procedure. We use a recent repair window of $W = 192$ hours and a forecaster context of $L_t = \min(8192, t)$ available hourly positions. Regression coefficients and VAR parameters are fitted on the same historical prefix. The recent observation mask determines which regression predictors are usable; it does not change the peer-selection rule or the fusion weight.

4.1 PREFIX-FITTED PEER REGRESSION

For each PM target, we select three other stations by absolute correlation with that target using jointly observed fitting-prefix values. The six selected PM channels augment the original eleven numerical variables. The peer identities are fixed before evaluation. For a missing target k at time u , let $\mathcal{S}_{uk}$ contain the other variables actually observed at that time. Estimated entries are never fed back as observations to the regression.

For each distinct pair $(k, \mathcal{S})$, let $\mathcal{V}_{k,\mathcal{S}}$ be prefix rows on which the target and all predictors in $\mathcal{S}$ are observed. At least 128 such rows are required. If support is insufficient, we remove the predictor with the smallest absolute prefix correlation with k until sufficient support is available. Correlation ties are resolved by the fixed variable order. With $n = |\mathcal{V}_{k,\mathcal{S}}|$, we fit

$$(\hat{\alpha}_{k,\mathcal{S}}, \hat{\beta}_{k,\mathcal{S}}) = \arg \min_{\alpha, \beta} \frac{1}{n} \sum_{u \in \mathcal{V}_{k,\mathcal{S}}} (z_{uk} - \alpha - z_{u,\mathcal{S}}^T \beta)^2 + \lambda \|\beta\|_2^2, \quad \lambda = 10^{-3}. \quad (1)$$

The intercept is unpenalized. Equation 1 has a regularized linear-system solution, cached for repeated predictor subsets. A missing value is estimated as $\hat{z}_{uk} = \hat{\alpha}_{k,\mathcal{S}} + z_{u,\mathcal{S}}^T \hat{\beta}_{k,\mathcal{S}}$. If no supported predictor remains, the fixed fallback uses supported linear interpolation, carry-forward values and, when no history anchor exists, the target’s prefix median. All fallback inputs precede t .

4.2 PROTECTED LONG-HISTORY CONSTRUCTION

Let Z^{long} be the standardized history on $[t - L_t, t)$. The current masked suffix replaces its last W positions before any repair, so extending the context cannot restore artificially hidden values. Define the target repair region $\mathcal{R}_t = \{(u, k) : t - W \leq u < t, k \in \mathcal{T}, M_{uk} = 0\}$. The forecaster receives

$$\tilde{Z}_{uj} = \begin{cases} \hat{z}_{uj}, & (u, j) \in \mathcal{R}_t, \\ Z_{uj}^{\text{long}}, & \text{otherwise.} \end{cases} \quad (2)$$

Table 1: Prediction procedure. Statistical fitting uses only the historical prefix; steps 2–6 use only information available before the origin.

	Input: prefix data, observed long history and mask, fixed peers, horizon H .
1	Fit prefix scales, supported peer-regression models and the stable VAR.
2	Extract the most recent W observations and construct the long input with its current mask.
3	Estimate each missing recent target from its observed predictors using Equation 1.
4	Insert these estimates only in $\mathcal{R}_t$ and query the frozen forecaster once.
5	Filter the original recent observations and recursively compute the VAR forecast.
6	Return the fixed mean in Equation 5, transformed to target units.

Equation 2 leaves all older gaps and all auxiliary gaps in their native form. The eleven local variables are retained. An appended peer with no observation in the current 192-hour window is omitted consistently from the compared model inputs; this decision remains fixed when extending the history. The model’s grouped multivariate interface then produces $P = F_H(\tilde{Z})$ for the target variables. Its original weights, internal normalization and attention computation remain unchanged.

4.3 STATISTICAL FORECAST AND FIXED FUSION

The statistical branch uses the standardized D -variable system

$$z_{u+1} = Az_u + b + \varepsilon_u, \quad \varepsilon_u \sim \mathcal{N}(0, Q). \quad (3)$$

We fit a first-order VAR on originally complete adjacent prefix rows, using centered ridge regression with penalty 10^{-3} . At least 128 complete transitions are required. The fitted transition matrix is multiplied by $\min(1, 0.99/\rho(A))$, where ρ denotes spectral radius, and b is recomputed from the transition-pair means. If S is the residual covariance, $Q = 0.95S + 0.05 \text{diag}(S) + 10^{-6}I$. The initial mean and covariance are estimated from complete prefix rows, with the same covariance regularization.

For each origin, this branch filters the *unrepaired* recent W observations. If m^-, C^- are the predicted state moments and $\mathcal{O}_u$ indexes the observed coordinates, its exact-observation update is

$$\begin{aligned} m_u &= m^- + C_{:, \mathcal{O}_u}^- (C_{\mathcal{O}_u, \mathcal{O}_u}^-)^{-1} (z_{u, \mathcal{O}_u} - m_{\mathcal{O}_u}^-), \\ C_u &= C^- - C_{:, \mathcal{O}_u}^- (C_{\mathcal{O}_u, \mathcal{O}_u}^-)^{-1} C_{\mathcal{O}_u, :}^- \end{aligned} \quad (4)$$

Empty observation sets leave the predicted moments unchanged. The observed coordinates are assigned their original values and zero conditional covariance. Between observations, Equation 3 propagates the moments. Starting from the final filtered state m_{t-1} , deterministic recursion $m_{t-1+h} = Am_{t-2+h} + b$ for $h = 1, \dots, H$ gives target forecast $G_h = (m_{t-1+h})_{\mathcal{T}}$. This Gaussian system is a working statistical approximation, with no assumption that it represents the true sensor process.

The returned forecast is

$$\hat{Y}_{hk} = \mu_k + s_k \hat{Z}_{hk}, \quad \hat{Z} = \frac{1}{2}P + \frac{1}{2}G. \quad (5)$$

Equation 5 uses the same weight for every station, target, horizon and missingness panel. There is no learned router, outcome-dependent threshold or per-panel choice of the repair algorithm. The VAR state in Equation 4 is inferred from observations rather than from the peer-regression repair, giving the two branches distinct statistical constructions under common pre-origin information.

Table 1 summarizes fitting and prediction for one origin. In the PM outage example, the missing target coordinates are estimated from reporting local and peer sensors, while both forecast branches retain their specified pre-origin observations.

4.4 PROPERTIES AND COMPUTATIONAL REQUIREMENTS

Preserved information. By construction, the repaired input agrees with the supplied input on every originally observed cell, every auxiliary cell and every position before $t - W$. This follows directly from the two cases in Equation 2. Extending the history before inserting the masked suffix also preserves artificial hiding. Prefix-fitted coefficients, current observed predictors and pre-origin

fallback values determine every inserted value. Consequently, the procedure is measurable with respect to the declared pre-origin information. This property concerns input validity; it is not an accuracy guarantee.

When fusion lowers squared error. For a fixed set of scored coordinates, write $e_P = P - Y^z$, $e_G = G - Y^z$, and let $\langle \cdot, \cdot \rangle_V$ denote the target-balanced average inner product on those coordinates. The established averaging identity gives

$$\|\tfrac{1}{2}(e_P + e_G)\|_V^2 = \tfrac{1}{2}\|e_P\|_V^2 + \tfrac{1}{2}\|e_G\|_V^2 - \tfrac{1}{4}\|P - G\|_V^2. \quad (6)$$

Equation 6 also holds after expectations or the same station aggregation. Fusion improves on P precisely when $\|e_G\|_V^2 + 2\langle e_P, e_G \rangle_V < 3\|e_P\|_V^2$. Large disagreement alone is insufficient if the statistical branch is inaccurate. This explains why the experimental comparison must retain G , P and their fixed mean separately. No analogous equality is assumed for MAE.

Fitting and inference cost. With N prefix rows and J distinct target–predictor subsets, direct regression fitting costs at most $O(J(ND^2 + D^3))$; cached coefficients are reused. Repair evaluation costs $O(WKD)$, and dense filtering costs at most $O(WD^3)$ plus $O(HD^2)$ for extrapolation. The primary procedure requires one frozen-model forward evaluation on the long history, together with these statistical operations. Its neural parameters receive no training updates. Multi-candidate prediction pools and LoRA training have additional costs; shared experimental caches do not remove their fitting or inference requirements.

5 EXPERIMENTAL EVALUATION

We evaluate the fixed procedure in Equation 5 on three Beijing panels, and examine statistical-component transfer on three HDB panels. All reported errors use one consistent result snapshot and the original future observation masks. The method and its application focus were selected retrospectively using these development results. Consequently, the experiments characterize performance on the observed cohorts and do not constitute independent confirmation of the selected method. The same primary definition is retained across the three Beijing panels.

5.1 DATA, INFORMATION AND METRICS

Beijing sensor forecasting. The Beijing Multi-Site Air Quality dataset contains hourly measurements from 12 stations between March 2013 and February 2017 (Chen, 2017). We predict PM2.5 and PM10 from eleven numerical station variables and six prefix-selected peer PM channels. All statistical fitting uses the first 7012 hours of each augmented station series. Evaluation origins follow the 60% temporal boundary, with the current 192-hour context entirely after that boundary. The primary and matched long-history controls use 8192 pre-origin hours. Related measurements use the provider’s aligned hourly clock; their availability before the origin is assumed, without a separate publication-delay model.

Natural-outage origins require both PM targets to have been continuously missing for at least six hours. Origins occur at outage ages $6 + 24k$ hours, with nonoverlapping 24-hour forecast intervals within a station. Each target must have at least twelve observed future values to be scored. The resulting panel contains 45 windows across 12 stations; 32 additional candidate origins lack sufficient observed future labels. This ascertainment can favor events that recover observations sooner. Artificial outages use four complete histories per station, selected at evenly spaced chronological positions, and hide only the last 24 hours of the two targets. The 138-window horizon-96 panel retains original target missingness and requires at least 48 observed future values per target. Table 2 summarizes these populations and the HDB component study.

HDB component-transfer population. The second source is Singapore HDB carpark availability (Housing & Development Board, 2026). Hourly snapshots cover 1 June–26 July 2025, yielding 1344 time positions and 2005 C-type carpark series. The first 336 hours supply fitting statistics and peer selection; the next 336 hours supply the development cohort. All eighteen prefix-eligible carparks with natural events in that development period are retained. The remaining two 336-hour intervals are unscored. Natural gaps, missing records, stale observations and collection failures are

Table 2: Evaluation populations. Windows are prediction tasks, and sites within a source are correlated. All panels are developmental. The HDB study evaluates available statistical-component variants, with no recorded result for the exact peer-regression primary.

Source	Panel	Sites	Windows	Horizon (h)
Beijing	Natural target outage	12	45	24
Beijing	Artificial target outage	12	48	24
Beijing	Original missing history	12	138	96
HDB	Natural availability gap	18	19	24
HDB	Artificial target outage	18	32	24
HDB	Native hourly grid	18	32	24

distinguished, with technical failures excluded from the natural-missingness interpretation. A target is accompanied by six prefix-selected peers. Long histories contain the available 349–586 pre-origin hours. The exact Beijing peer-regression method has no recorded HDB evaluation; the Gaussian repair, fixed fusion and native controls permit a narrower component-transfer comparison.

Evaluation metric. Let $n_k = \sum_h V_{hk}$ and let s_k be the original-observation prefix scale. We compute

$$\begin{aligned} \text{MAE}(\hat{Y}, Y) &= \frac{1}{K} \sum_k \frac{1}{n_k} \sum_{h: V_{hk}=1} \frac{|\hat{Y}_{hk} - Y_{hk}|}{s_k}, \\ \text{MSE}(\hat{Y}, Y) &= \frac{1}{K} \sum_k \frac{1}{n_k} \sum_{h: V_{hk}=1} \frac{(\hat{Y}_{hk} - Y_{hk})^2}{s_k^2}. \end{aligned} \quad (7)$$

Equation 7 averages targets within a window, then windows within each site, and finally sites with equal weights. Missing future values are never completed to manufacture labels. MAE is primary, and both metrics are retained for every comparison. Relative change is $100(\text{method}/\text{reference} - 1)$, so negative values denote lower error. We report paired site directions and leave-one-site-out ranges as descriptive sensitivity analyses, without treating them as confidence intervals or converting repeated development comparisons into significance claims.

5.2 COMPARATORS AND FITTING BUDGETS

The frozen backbone is the same 120M-parameter Chronos-2 checkpoint throughout (Ansari et al., 2025). Native controls retain missing values and use either 192 hours or the available long context. Separate controls retain only the target variables, providing an explicit input-scope comparison. VAR uses the same fitted statistical branch as the primary. Native-plus-VAR controls hold both the history and fusion weight fixed while omitting target repair.

Repair controls include LOCF, linear interpolation, seasonal copying, multivariate KNN, SAITS, TimeMixer++ and MoTM. The neural imputers use two shared 17-variable Beijing fits, each trained on 512 masked prefix windows for 50 epochs, with batch size 16 and seed zero. MoTM retains its released weights and its fixed observed-context adaptation procedure. Short-history comparisons include each target-only repair and the mean/median of the seven repaired predictions plus native prediction. Additional all-variable repairs and statistical regression variants are retained in the supplement. A fixed Gaussian control estimates prefix means and regularized covariance and applies the conditional mean independently at each time. The same long-history and VAR-fusion factors are applied to Gaussian, KNN and ridge repairs.

Two LoRA controls add 1,206,912 parameters to 97 Chronos-2 projections, with rank eight and scaling parameter sixteen. Natural-history and corruption-trained versions share initialization, source labels, update order and 1728 single-task updates, split equally between Beijing and HDB. They use standardized smooth L1 loss with parameter 0.01, AdamW learning rate 10^{-5} , weight decay 0.01 and gradient-norm limit one. Evaluation uses the final update. These adapted controls have a different training budget from the statistical primary; their stronger results are still retained. Appendix A gives the source boundaries and additional implementation details.

Table 3: Beijing results. Each row has one fixed definition across panels. Long histories contain 8192 hours; unqualified inputs include the available local and peer variables. “Targets only” changes the variables supplied to the forecaster. Values are standardized MAE/MSE; lower is better. Bold marks the best value in each displayed column.

Method	Natural, $H = 24$		Artificial, $H = 24$		Original, $H = 96$	
	MAE	MSE	MAE	MSE	MAE	MSE
Native, 192 h	0.5302	0.4359	0.8300	1.2501	0.8633	1.6678
Native, long	0.4848	0.3593	0.6665	0.8567	0.7965	1.4456
Native, long, targets only	0.5351	0.4406	0.6786	1.0608	0.7898	1.4165
VAR	0.4336	0.3037	0.6628	0.8248	0.8740	1.3634
Native, long + VAR	0.4128	0.2638	0.6354	0.7795	0.8098	1.3269
Native, long, targets + VAR	0.4347	0.2914	0.6170	0.7453	0.8052	1.3130
Target KNN, 192 h + VAR	0.3970	0.2644	0.6648	0.9118	0.8342	1.4266
Gaussian repair, long	0.3918	0.2524	0.6289	0.8660	0.7966	1.4498
Gaussian repair, long + VAR	0.3882	0.2451	0.6310	0.8110	0.8100	1.3290
KNN repair, long + VAR	0.3887	0.2441	0.6353	0.8286	0.8097	1.3293
Local ridge, long, targets only	0.5156	0.4371	0.5076	0.5421	0.7963	1.4325
Corruption-trained LoRA, 192 h	0.5067	0.4199	0.8151	1.3787	0.8573	1.5938
Peer Repair and Fusion	0.3881	0.2447	0.6311	0.8118	0.8099	1.3289

5.3 MAIN RESULTS ON NATURAL SENSOR OUTAGES

Table 3 compares fixed methods across all three Beijing panels. On natural outages, Peer Repair and Fusion reaches MAE/MSE 0.388053/0.244688. Relative to the long native forecaster, the reductions are 19.96% and 31.90%. The matched native-plus-VAR comparison isolates the effect of adding target repair under the same long history and statistical forecast: MAE/MSE decrease from 0.412755/0.263833 by 5.98%/7.26%. Against short target-KNN repair with the same VAR fusion, reductions are 2.25%/7.46%.

The nearest repair controls substantially narrow the interpretation. Gaussian repair with the same long history and VAR obtains 0.388168/0.245143, only 0.03%/0.19% above the primary. The primary wins both metrics at four of twelve stations against this Gaussian control, and its leave-one-station-out MAE direction can reverse. Long KNN repair with VAR obtains 0.388676/0.244071: its MAE is higher, but its MSE is lower. The data therefore support a useful class of simple repair-and-fusion procedures in this cohort, with no established substantial superiority of peer ridge over Gaussian repair.

The improvement over native-plus-VAR is less sensitive to station composition. The primary improves MAE at nine stations and MSE at eight. After omitting each station in turn, its relative changes range from -7.52% to -4.91% for MAE and from -9.69% to -5.40% for MSE. Against short KNN-plus-VAR, the corresponding ranges are $[-3.96\%, -1.42\%]$ and $[-9.24\%, -6.17\%]$. These ranges describe the observed station population; its twelve stations remain one related source. Appendix B reports the individual station results.

5.4 HISTORY, REPAIR AND FUSION EFFECTS

Figure 2 crosses two history lengths, target repair and fixed VAR fusion on the same natural-outage windows. Extending native history from 192 to 8192 hours changes MAE/MSE from 0.530193/0.435904 to 0.484839/0.359322, a reduction of 8.55%/17.57%. Thus a short native input is an insufficient reference for a method that can read the longer history.

At the long history, peer repair alone obtains 0.394116/0.254031, while its fixed fusion obtains 0.388053/0.244688. The repair operation already accounts for much of the natural-outage improvement, and fusion adds a smaller increment. Gaussian repair alone also performs well at 0.391787/0.252425, reducing MAE/MSE by 19.19%/29.75% relative to the same long native input. These comparisons separate additional historical information, target repair and statistical forecast combination. They do not attribute all of the gain to a learned decision mechanism.

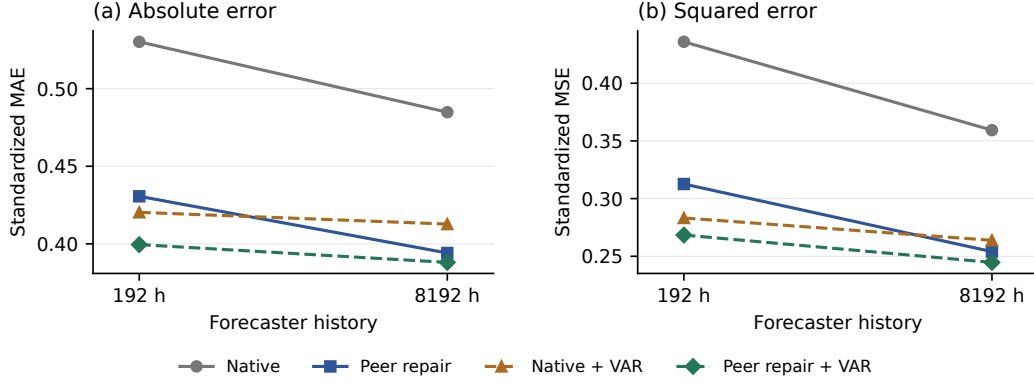


Figure 2: Component comparison on the 45 natural-outage Beijing windows. Each line retains its repair and fusion rule while changing only the history supplied to Chronos-2. The repair region and VAR filtering window remain 192 hours, and prefix-fitted parameters are unchanged. The plot reports observed station-macro errors; it does not show independent replicate uncertainty.

Artificial outages and longer horizons produce different rankings. On artificial target outages, the primary reaches 0.631104/0.811784. Local ridge repair followed by long target-only forecasting is substantially better at 0.507640/0.542087. Relative to long native-plus-VAR, the primary has slightly lower MAE and higher MSE. At horizon 96, the primary obtains 0.809889/1.328908, while long target-only native forecasting obtains the lower MAE 0.789801 and its VAR fusion obtains the lower MSE 1.313030. The primary remains fixed in these comparisons; the stronger row in each panel is not substituted into its definition.

5.5 TRANSFER OF STATISTICAL COMPONENTS TO HDB

Table 4 reports the three HDB panels without filling in an unavailable result for the peer-regression primary. Gaussian target repair and its fixed VAR fusion instantiate the available statistical components with the same fitting-only data boundary. On natural gaps, long native target-only forecasting obtains 1.178178/3.887765. Long Gaussian repair with peers obtains 1.434379/4.706353, and VAR fusion further raises MAE to 1.546516. Even the best-MAE evaluated long repair, target-only median repair at 1.263786/4.012167, remains 7.27%/3.20% above the native target-only control. Two additional query-mask controls retain the long native history while restricting unknown auxiliary-future attention keys along time or along both time and group dimensions. They consume no measured future inputs and are retained for their competitive results.

On HDB artificial outages, corruption-trained LoRA gives 0.643229/1.120020, outperforming the 336-hour native control by 4.08%/7.42%. This adapted model does not exceed the strongest native control on natural gaps. On the native-grid panel, Gaussian repair without fusion is close to long native forecasting, while the VAR branch has considerably higher error. The outcome is consistent with the component-error requirement in Equation 6; it does not identify a single cause of the cross-source difference. The HDB results constrain claims about transferring a Beijing repair-and-fusion design to other applications.

6 DISCUSSION

The main empirical result is a reduction in forecasting error after naturally missing PM histories when related observations, target-only statistical repair and a long model context are combined. Equal fusion adds a smaller improvement, and a Gaussian replacement for peer ridge performs nearly identically. This places the result at the level of a concrete applied forecasting procedure. Ridge regression, Gaussian conditioning and forecast averaging are established components, and the present evidence does not isolate a new statistical principle from their combination.

Table 4: HDB component-transfer results. Long histories use all available pre-origin values up to the fixed context limit. The exact peer-regression primary is unmeasured on HDB; Gaussian rows are separately defined component variants. Bold marks the lowest displayed error.

Method	Natural gaps		Artificial outage		Native grid	
	MAE	MSE	MAE	MSE	MAE	MSE
Native, 192 h	1.2617	4.1144	0.6863	1.2566	0.4657	0.6436
Native, long	1.2006	3.9136	0.6796	1.3017	0.4488	0.6405
Native, long, target only	1.1782	3.8878	0.6789	1.3722	0.4521	0.6482
VAR	1.7269	4.9909	1.2949	3.1104	1.1826	2.7344
Native, long + VAR	1.4071	3.9467	0.8507	1.3853	0.7319	1.0201
Gaussian repair, long	1.4344	4.7064	0.9145	2.1494	0.4483	0.6424
Gaussian repair, long + VAR	1.5465	4.5873	1.0598	2.3915	0.7324	1.0205
Median repair, long, target only	1.2638	4.0122	1.2083	2.3544	0.4528	0.6495
Median repair, long, with peers	1.2662	3.9810	1.1726	2.2521	0.4477	0.6380
Auxiliary-query mask, time	1.1938	3.9026	0.6689	1.2575	0.4468	0.6390
Auxiliary-query mask, time/group	1.1827	3.8757	0.6709	1.3002	0.4509	0.6480
Nine-candidate median, 192 h	1.2435	4.0190	0.8211	1.6246	0.4727	0.6483
Corruption-trained LoRA, 192 h	1.2330	3.9595	0.6432	1.1200	0.4690	0.6497

The application scope and primary method were chosen using repeatedly examined development data. Missing future labels restrict scoring to observed outcomes, and the event eligibility conditions exclude prolonged gaps with insufficient future support. Source, calendar period, label observability and missingness differ across the natural and artificial panels, preventing a causal attribution of their ranking differences to missingness alone. The exact primary has been evaluated on one source and one frozen backbone; the HDB component study is smaller and does not evaluate that exact procedure. Exclusion of all evaluation sources from foundation-model pretraining has not been established. Independent data with frozen eligibility and fitting rules are necessary to assess generalization of the selected procedure.

7 CONCLUSION

Peer Repair and Fusion combines prefix-fitted peer regression, protected long-history input and fixed statistical forecast fusion. On the studied Beijing natural-outage cohort, it reduces MAE/MSE by 5.98%/7.26% relative to native prediction with the same long history and VAR branch. Comparable Gaussian repair, stronger controls on artificial outages and longer horizons, and unfavorable HDB component transfer delimit the result. The resulting method is fully specified and executable; its predictive advantage is currently supported for the observed natural-outage sensor cohort.

REFERENCES

- Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025. URL <https://arxiv.org/abs/2510.15821>.
- J. M. Bates and C. W. J. Granger. The combination of forecasts. *Operational Research Quarterly*, 20(4):451–468, 1969. doi: 10.1057/jors.1969.103. URL <https://www.tandfonline.com/doi/abs/10.1057/jors.1969.103>.
- Song Chen. Beijing multi-site air quality. UCI Machine Learning Repository, 2017. URL <https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data>.

- Wenjie Du, David Côté, and Yan Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 219:119619, 2023. doi: 10.1016/j.eswa.2023.119619. URL <https://doi.org/10.1016/j.eswa.2023.119619>.
- Wenjie Du, Jun Wang, Linglong Qian, Yiyuan Yang, Zina Ibrahim, Fanxing Liu, Zepu Wang, Haoxin Liu, Zhiyuan Zhao, Yingjie Zhou, Wenjia Wang, Kaize Ding, Yuxuan Liang, B. Aditya Prakash, and Qingsong Wen. TSI-Bench: Benchmarking time series imputation. *arXiv preprint arXiv:2406.12747*, 2024. URL <https://arxiv.org/abs/2406.12747>.
- Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634. URL <https://www.tandfonline.com/doi/abs/10.1080/00401706.1970.10488634>.
- Housing & Development Board. Carpark availability. data.gov.sg, 2026. URL https://data.gov.sg/datasets/d_ca933a644e55d34fe21f28b8052fac63/view. Hourly historical snapshots for June–July 2025.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021. URL <https://arxiv.org/abs/2106.09685v2>.
- Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems*, volume 7, pp. 231–238, 1994. URL https://proceedings.neurips.cc/paper_files/paper/1994/file/b8c37e33defde51cf91e1e03e51657da-Paper.pdf.
- Marine Le Morvan, Julie Josse, Erwan Scornet, and Gael Varoquaux. What’s a good imputation to predict with missing values? In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL <https://proceedings.neurips.cc/paper/2021/hash/5fe8fdc79ce292c39c5f209d734b7206-Abstract.html>.
- Etienne Le Naour, Tahar Nabil, and Ghislain Agoua. MoTM: Towards a foundation model for time series imputation based on continuous modeling. *arXiv preprint arXiv:2507.13207*, 2025. URL <https://arxiv.org/abs/2507.13207>.
- Jing Peng, Meiqi Yang, Qiong Zhang, and Xiaoxiao Li. S4M: S4 for multivariate time series forecasting with missing values. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/7b2f0758334389b8ad0665a9bd165463-Abstract-Conference.html.
- Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenze Lin, Shengtong Ju, Zhixuan Chu, and Ming Jin. TimeMixer++: A general time series pattern machine for universal predictive analysis. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=1CLzLXSFNn>.

A FITTING DETAILS AND COMPARISON DEFINITIONS

Primary procedure. The eleven Beijing variables are PM2.5, PM10, SO2, NO2, CO, O3, temperature, pressure, dew point, rainfall and wind speed. Wind direction and calendar identifiers are excluded. Three other stations are selected for each PM target by absolute jointly observed prefix correlation. All current target estimates use observed predictors, with the 128-row support threshold and 10^{-3} ridge penalty in Equation 1. The forecaster reads prefix-standardized float32 inputs; output scoring retains the same prefix units. Original observations are copied directly into every model-input representation. No additional normalization, attention mask or learned embedding is introduced by the primary procedure.

The VAR fits originally complete adjacent rows in the same first-7012-hour prefix. The twelve stations provide 4052–5990 complete transitions each. Spectral-radius stabilization, covariance regularization, exact-observation filtering and equal output fusion are fixed as described in Section 4. The forecaster reads the long history, whereas the regression repair and VAR state estimation use the recent 192-hour window. This asymmetric allocation is part of the method definition and is shared by the matched controls.

Gaussian and local-regression controls. Let m and o index missing and observed coordinates at one time. The static Gaussian repair is $\mu_m + \Sigma_{mo}\Sigma_{oo}^{-1}(x_o - \mu_o)$, with prefix-fitted moments and the same covariance shrinkage as the statistical system. It conditions independently at each time, without VAR transitions. Local ridge restricts regression predictors to the original eleven variables. In the “targets only” forecasting control, the repair may still have used local or peer observations; only the variables supplied directly to the frozen forecaster change. Consequently, target-only forecasting is not a no-covariate-information experiment.

Neural imputation. SAITS and TimeMixer++ use shared fits on 512 windows of length 192 drawn from the twelve augmented training prefixes, with stride 24. Training masks span random points, independent blocks, synchronous blocks, staggered correlated blocks, value-dependent missingness and mixed outages, with rates 0.1–0.5 and seeds 1101–1103. Each model trains for 50 epochs with batch size sixteen and initialization seed zero. SAITS uses two layers, model dimension 64, feed-forward dimension 128 and four heads. The implementations use their imputation objectives and are not jointly optimized through Chronos-2. MoTM retains its released neural parameters and the fixed context-adaptation settings, including ridge coefficient 0.5. All compared repairs preserve original observations and share the fixed finite-value fallback where needed.

Low-rank adaptation. The two LoRA controls augment every self-attention query, key, value and output projection and the output-patch projection. The down-projection uses Kaiming-uniform initialization and the up-projection is zero, with seed 7402. Beijing supplies 864 source tasks from 96 histories; their training futures end no later than hour 7012 and their preprocessing prefix ends at hour 3506. HDB uses only the first 336 hours and anchors 192, 216, 240, 264, 288 and 312. HDB preprocessing statistics and peers use that available 336-hour training set, rather than a strictly preceding subset for each source anchor. Both adaptations receive the same future labels. Corruption training adds fixed target-only, target-plus-peer and all-variable outages of ages 6, 24 and 54 hours while retaining native missingness. The source pool has 1836 tasks from 204 histories; a fixed source-balanced order selects 864 updates per source, with 1728 updates in total. No current evaluation future is used for fitting.

Table 5 retains individual imputers and prediction-pool comparisons, with no substitution of a method chosen from each panel’s future outcomes. The complete supplementary CSV includes 1275 panel–method rows: 266 fixed output definitions on each Beijing panel and 159 on each HDB panel. Many outputs vary history length, variable scope or a fixed combination and are not distinct learning algorithms. The accompanying site-level CSV permits reconstruction of each reported macro-average.

B STATION-LEVEL NATURAL-OUTAGE RESULTS

Table 6 reports the primary method at each Beijing station and its paired relative change from the long-native-plus-VAR control. This control uses the same auxiliary information, long history, VAR

Table 5: Additional Beijing comparisons. The seven imputation rows repair only the two target variables in the 192-hour input; auxiliary values and missingness remain unchanged. The eight-candidate mean and median include those seven forecasts and native prediction. Regression and LoRA controls use the input definitions described in the text. Lower standardized MAE/MSE is better; bold marks the best displayed value.

Method	Natural, $H = 24$		Artificial, $H = 24$		Original, $H = 96$	
	MAE	MSE	MAE	MSE	MAE	MSE
LOCF	0.6548	0.6763	1.0132	1.7746	0.8844	1.7592
Linear interpolation	0.6525	0.6677	1.0132	1.7746	0.8856	1.7636
Seasonal lag	0.5515	0.5482	0.8531	1.4398	0.8838	1.7505
Multivariate KNN	0.4370	0.3171	0.7015	1.0882	0.8766	1.7355
SAITS	0.4536	0.3248	0.6802	0.9918	0.8806	1.7555
TimeMixer++	0.5272	0.4876	0.8922	1.4330	0.8890	1.7696
MoTM	0.6391	0.6449	0.9486	1.6389	0.8872	1.7604
Eight-candidate mean	0.5156	0.3950	0.8052	1.1213	0.8786	1.7309
Eight-candidate median	0.4981	0.3751	0.7830	1.1428	0.8799	1.7457
Peer ridge	0.4306	0.3126	0.6921	1.0691	0.8738	1.7218
Peer ridge + AR residual	0.4381	0.3178	0.6922	1.0676	0.8806	1.7509
Peer ridge + linear residual	0.4855	0.3728	0.7370	1.1795	0.8811	1.7511
Natural-history LoRA	0.5226	0.4537	0.8506	1.4419	0.8722	1.6702
Corruption-trained LoRA	0.5067	0.4199	0.8151	1.3787	0.8573	1.5938

parameters and fusion coefficient; it omits peer-regression target repair. Site scores average the available windows within the site, followed by an equal-site macro-average. A site with more eligible windows therefore receives no additional weight in the reported overall result.

Table 6: Natural-outage station results for Peer Repair and Fusion. Relative changes compare with long native prediction plus VAR. Negative changes denote lower error. The twelve stations are related observations from one source.

Station	MAE	MSE	Δ MAE (%)	Δ MSE (%)
Aotizhongxin	0.2974	0.0962	-19.34	-33.57
Changping	0.4616	0.3500	-7.78	-15.28
Dingling	0.3868	0.1930	-3.26	-15.54
Dongsi	0.3756	0.2278	-8.89	-7.63
Guanyuan	0.4351	0.2516	+12.08	+27.37
Gucheng	0.5523	0.5210	-6.18	-4.55
Huairou	0.3603	0.2024	-15.70	-17.49
Nongzhanguan	0.2624	0.1322	-5.78	+2.46
Shunyi	0.4823	0.4131	+2.93	+10.97
Tiantan	0.3471	0.1761	+1.11	+2.98
Wanliu	0.3298	0.1603	-8.06	-13.51
Wanshouxigong	0.3660	0.2126	-12.54	-25.85

C HDB OBSERVATION SEMANTICS AND RESERVED PERIODS

Hourly HDB values represent the latest eligible report of available C-type spaces. Each request uses the grid time minus one minute; snapshots and carpark updates must fall within the preceding hour. Raw values, capacities, update times and record ages are retained, with separate status for absent, stale, conflicting or malformed records and technical failures. An eligible report need not be a physical measurement made exactly on the hour.

The 1344-hour collection contains 2,694,720 carpark-hour cells. Fitting uses $[0, 336)$ and development uses $[336, 672)$; $[672, 1008)$ and $[1008, 1344)$ remain unscored. Nineteen natural-gap tasks from the eighteen eligible carparks support the reported component-transfer comparison. Their small number and future-observation requirements constrain inference to this cohort.